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Palaeomagnetism of the Early Carboniferous Deer Lake Group, western Newfoundland: no evidence for mid-Carboniferous displacement of "Acadia"

E. Irving [1] and D.F. Strong [2]

[1] *Pacific Geoscience Centre, Sidney, B.C. V8L 4B2 (Canada)*
[2] *Department of Earth Sciences, Memorial University, St. John's, Newfoundland A1B 3X5 (Canada)*

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Early Carboniferous (Viséan to possibly earliest Namurian) sedimentary rocks of the Deer Lake Group of western Newfoundland rest unconformably on Grenvillian basement rocks of the Canadian Shield which form the western border of the Early Palaeozoic Appalachian orogen. In addition to magnetically soft magnetizations directed along the present field, three families of magnetization directions are found. Two of them (referred to as N (north) and S (south)) are antiparallel and prefolding, and were probably acquired during the depositional process. N and S are roughly of equal frequency. They have a mean direction irrespective of sign of 0.7°, −35.2°, $k = 40$, $\alpha_{95} = 8.9°$, and a palaeopole 21.5°N, 121.8°E (10.3°, 6.0°) corresponding to a palaeolatitude of 20 ± 6°S. This agrees with the palaeolatitude (17 ± 5°S) determined from the somewhat older Early Carboniferous (Tournaisian) Terenceville Formation of the Avalon Platform on the eastern side of the Appalachian orogen in Newfoundland. The third magnetization, referred to as H (roughly horizontal), has a mean direction 156.8°, −13.3°, $k = 37$, $\alpha_{95} = 10.1°$, and a palaeopole 45.4°N, 140.3°E (10.3°, 5.3°) corresponding to a palaeolatitude of 7 ± 4°N; we interpret this to be an early Kiaman (latest Carboniferous to early Permian) overprint probably acquired chemically. The palaeolatitude determined from the H overprint agrees with that determined from Early Carboniferous rocks of cratonic North America west of the Appalachians. Therefore, we argue, Early Carboniferous palaeofield determinations for cratonic North America have been strongly biased by unremoved Kiaman overprints. Because of this, and because of the good agreement between Early Carboniferous palaeolatitudes obtained from opposite margins of the Appalachian orogen, we suggest that there is, at present, no palaeomagnetic evidence for the previously proposed 1500 km displacement from the south of an eastern portion of the Appalachians ("Acadia") relative to cratonic North America during the Carboniferous.

1. Introduction and geology

This work was undertaken to determine the palaeolatitude at which the Early Carboniferous Deer Lake Group was deposited. The Deer Lake Basin is filled with Carboniferous sedimentary rocks [1–4]. They are mainly detrital and were laid down in fresh-water. Along the northwest margin they rest unconformably on Precambrian (Grenville Structural Province of the Canadian Shield) basement gneisses, a particularly good exposure being at Taylors Brook [1]. Elsewhere contacts are faulted. Boreholes through the Deer Lake Group penetrate basement gneisses [1]. Basal conglomerates of the group contain Grenvillian clasts. The basement forms the Early Palaeozoic platform of western Newfoundland and it is directly tied across the Straits of Belle Isle to the Grenville Province of the mainland by early Cambrian volcanics and sediments. There have long been discussions of possible anticlockwise rotation of Newfoundland by 10 or 15° in the later Phanerozoic [5], and although this is presently an open question there is no possibility that the Deer Lake

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basin and rocks west of it have been displaced laterally relative to the craton by more than a hundred kilometres or so. Hence a determination of palaeolatitude for the Deer Lake Group provides a check on previous determinations from Early Carboniferous rocks of cratonic North America. This is important because Kent and Opdyke [6] followed by others [7,8] have argued that data, mainly from the Maritime Provinces of Canada ("Acadia"), differ significantly from the cratonic reference field and hence that Acadia was about 1500 km south relative to the craton in the Devonian and that it moved to its present position during the Carboniferous. Our study therefore is a test of the Kent-Opdyke hypothesis. If the palaeolatitude of the Deer Lake Group should turn out to be the same as that generally accepted for the craton, their hypothesis would be supported. If it does not, then the accuracy of the cratonic reference field, and hence the evidence for Carboniferous motion, is called in question. That some strike-slip faulting [9] occurred in the later Palaeozoic, partly perhaps in the Carboniferous, is not questioned. The point at issue is whether or not the very large sinistral measurement invoked above has occurred.

Hyde [1] subdivided the deposits of the Deer Lake basin (Fig. 1) into the Tournaisian Anguille Group, the Wigwam Brook and Wetstone Point

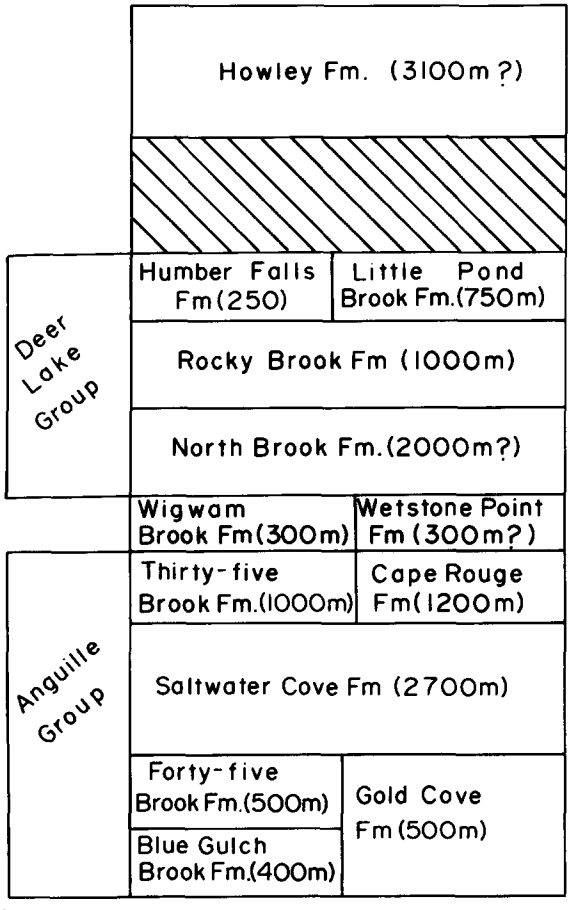


Fig. 1. Stratigraphic column in Deer Lake Basin compiled from references 1–4. The Anguille Group is Tournaisian. The Deer Lake Group is Viséan, possibly early Namurian at the top. The Howley Formation is Westphalian.

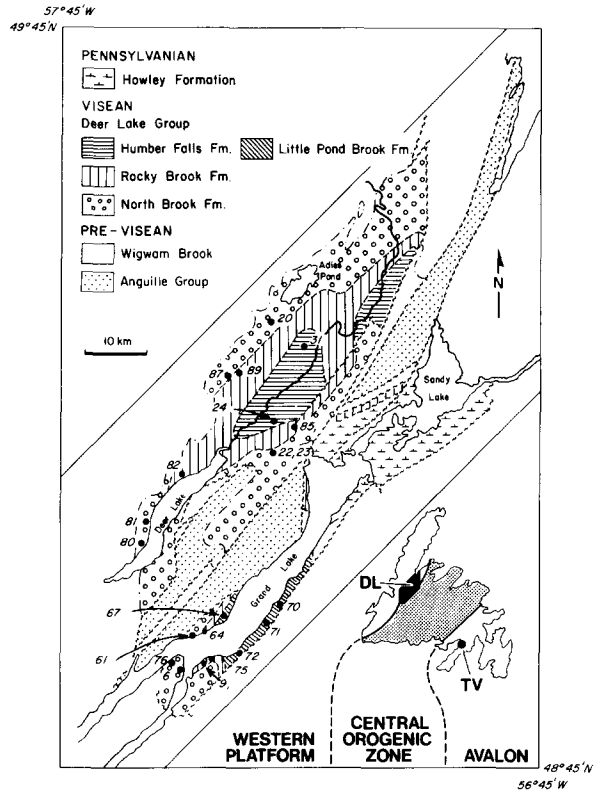


Fig. 2. Geological map of the Deer Lake Basin, simplified from Belt [4]. The collecting sites are coded by arbitrary numbers which have no geological significance. See Table 1. 61 and 64 were collected in error from slump beds and were not studied. 31 in very coarse-grained sandstone gave no coherent results. The inset shows the eastern (Avalon) and western Precambrian platforms and the intervening Palaeozoic orogenic belt comprised of the Dunnage and Gander Zones.

Formations, the unconformably overlying predominantly Viséan Deer Lake Group, and the Westphalian Howley Group (Figs. 1 and 2). Of the Early Carboniferous rocks, only the Deer Lake Group is sufficiently well-exposed and undeformed for palaeomagnetic study. It has four units.

The North Brook Formation, the lowermost unit, reaches a thickness of more than 2000 m. It consists mainly of red to grey arkosic sandstone and conglomerate, with minor red to grey siltstone and pink to grey micritic limestone. Finer-grained rocks are found towards the centre and top of the basin. Conglomerates occur towards the base and margins. Belt [4] and Hyde [2] interpret the rocks as fluvial and alluvial fan deposits. Except for calcified and silicified fragments of wood, the unit is unfossiliferous.

The Rocky Brook Formation conformably overlies the North Brook Formation, and its lowermost parts intertongue with facies equivalents of the North Brook. The formation attains a thickness of 1000 m, but 500–600 m is more typical. Fish remains, plant scraps, algal growths, calcareous worm tubes, and rare bivalves occur. Spores indicate a Viséan age [1]. Hyde [1] subdivided the formation into a lower member consisting of red calcareous siltstone, grey to green siltstones and mudstones, calcareous dolostones and dolomitic limestones, and an upper member which lacks redbeds and is dominated by grey, green, and black mudstones and siltstones, with carbonates. He considers the unit to be a mixed lacustrine-fluvial deposit, with numerous periods of subaerial exposure during deposition of the lower member, and an entirely subaqueous regime for the upper.

The Humber Falls Formation has a maximum thickness of 250 m, and overlies the Rocky Brook Formation, possibly disconformably. It consists of grey to reddish conglomerates and arkosic sandstones, and red to grey siltstones and mudstones. It was interpreted as an alluvial-fan to fluvial deposit by Belt [4], although Hyde [2] considers alluvial-fan deposition to be minor.

The Little Pond Brook Formation, maximum thickness of 750 m, is found only along the southeastern margin of the Deer Lake basin (Fig. 2), and is thought to be a lateral equivalent of the Humber Falls Formation. Spore assemblages in both units suggest a Viséan to possibly early Namurian age [1]. It has two informal members [3]. One, on the east side of Grand Lake, consists of grey to red clastics, rich in plant debris. A second, on the western shore of Grand Lake, gradationally overlies the Rocky Brook Formation, and consists of grey coarse-grained sandstone and pebble conglomerate interbedded with red fine-grained sandstone and siltstone. The Little Pond Formation has been interpreted as fluvial in origin [3].

The Deer Lake Group is displayed in two northeast-trending synclines, the Humber syncline in the west and the Grand Lake syncline (inferred, as its axial trace is beneath the lake) in the east (Fig. 2). Bedding attitudes vary from sub-horizontal to overturned. Plunge of folds is such that corrections would amount to less than 2° and, because this is small compared to errors in the palaeomagnetic determinations, no plunge corrections have been made. The fact that bedding dips are steepest near faults has led to the suggestion [2] that folding of the Deer Lake Group was the result of dip-slip movement along major northeasterly-trending faults. These faults controlled the overall form and sedimentation of the basin and may have been active, with strike-slip movement along them, during the deposition and deformation of the older Anguille Group.

2. Procedures

Oriented hand-samples were collected in 1981 and 1982 with emphasis on fine-grained sandstones and siltstones. Access is difficult and exposures are few and often weathered. Collecting sites (Fig. 2, Table 1) are generally along lake shores or the beds of brooks, accessible only at times of low water. Hence areal and stratigraphic coverage are unavoidably uneven. From each hand-sample up to 5 short cores were taken, and from these one, less commonly two, specimens were cut. Many samples broke during preparation or acid treatment. Useable specimens were obtained from 18 collecting sites (30 hand-samples, 116 cores, 123 specimens). Stratigraphically they span about 2000 m.

TABLE 1

Collecting site location and bedding attitude

Site	Location		DDA°/D°
	lat.(N)	long.(W)	
6	48°53.1′	57°31.0′	020, 15
9	48°53.5′	57°29.5′	300, 30
13	48°57.5′	57°21.0′	240, 20
20	49°22.0′	57°21.0′	130, 05
22	49°11.0′	57°23.5′	010, 27
24	49°13.5′	57°20.5′	315, 30
67	48°56.2′	57°26.2′	114, 95 [a]
70	48°57.9′	57°20.2′	306, 46
71	48°57.0′	57°21.7′	260, 25
72	48°53.8′	57°27.1′	067, 26
75	48°53.8′	57°28.7′	033, 26
76	48°53.5′	57°33.6′	040, 22
80	49°04.9′	57°36.9′	143, 11
81	49°05.7′	57°36.8′	110, 10
82	49°09.2′	57°32.5′	135, 08
85	49°13.2′	57°17.5′	340, 45
87	49°16.5′	57°27.5′	135, 05
89	49°17.8′	57°25.7′	120, 07

DDA° is the azimuth of down-dip direction, D° is the angle of dip. The site numbers are arbitrary field numbers and have no stratigraphic significance.

[a] Overturned.

All specimens have been subjected to thermal demagnetization (using a Schonstedt thermal demagnetizer) or acid (8*N* HCl) leaching, as described by Park [10]. Generally about 14 steps (never fewer than 7) were taken, sufficient to allow the various magnetizations to be separated in most specimens. Magnetizations were recognized from linear segments on orthogonal plots [11,12]. The presence of magnetizations of lower resistance (stability) to acid leaching or to heat is shown by segments that do not converge to the origin. The directions of these less stable magnetizations have been determined by vector subtraction, that is by determining the magnetization removed along linear segments. Magnetizations of highest stability give linear segments which converge to the origin (true end-point magnetizations) and their directions have been determined by averaging directions observed along the segment or by choosing a typical value along the segment. Commonly, however, magnetizations of the highest stability are very small in magnitude and tend to be obscured by experimental noise, and no linear segment can be accurately determined for them. In these instances only magnetizations of lower stability can be determined.

3. Examples of multi-component magnetizations

The first example, from the Little Pond Brook Formation (collecting site 67), shows a dominant magnetization pointing upwards and to the north-northeast. The direction is little changed during thermal demagnetization, but from 600 to 650° there is a tendency for the direction to shift toward the south (Fig. 3a). The orthogonal plots have two segments. The magnetization removed by heating to 100°C has a direction 352, +72 relative to present horizontal, which is negligibly different from the direction of the present earth's field (*PEF*) in western Newfoundland (349, +73). The second marches approximately, but not exactly, to the origin and corresponds to the removal of a magnetization (designated *N*) with a northerly direction and upward inclination (027, −47) relative to bedding. The fact that the second segment does not converge exactly to the origin shows that a third high temperature magnetization is present. This third component is also revealed by acid leading (Fig. 3b). First a small *PEF* component is removed, then a large *N* magnetization is apparent after immersion up to about 100 hours, but again the orthogonal plots do not march to the origin. For immersion longer than 200 hours an "apparent" end-point is reached but the magnetization remaining after 600 hours could still be composite. A small acid-resistant southerly-directed component is present, but it has not been accurately isolated.

A second example, this time from the Humber Falls Formation, is shown in Fig. 4. The thermal decay curve (Fig. 4, left-hand side) has a two-step form, the directions with respect to present horizontal being directed to the south. There is an "apparent" end-point with an inclination of about −13°. The orthogonal plots show a short *PEF* segment followed by a much longer one which converges approximately but not exactly to the origin, so a third magnetization is present. The

TABLE 2

Results by sites

Site	Formation	Magnetization	Reference plane	c, s	$D°, I°$	k	α_{95}°
06	NB	$S(N)$	b	7, 8	171, 37	48	8
06	NB	H	h	2, 3	175, −04	166	10
09	RB	S	b	5, 8	177, 34	21	13
09	RB	N	b	1, 1	015, −24	–	–
09	RB	$N+S$	b	6, 9	179, 33	21	12
09	RB	H	h	3, 3	178, −16	14	–
13	LPB	N	b	1, 1	351, −27	–	–
20	NB	H	h	2, 2	163, −54	149	–
22	NB	S	b	2, 2	200, 35	79	–
22	NB	N	b	2, 2	039, −34	29	–
22	NB	$N+S$	b	4, 4	209, 35	36	16
24	HF	S	b	2, 2	189, 51	17	–
24	HF	$H(S)$	h	8, 9	173, −16	84	6
67	LPB	$N(S)$	b	20, 23	014, −41	70	4
70	LPB	N	b	2, 2	036, −40	11	–
70	LPB	H	h	3, 3	167, −28	1070	–
71	LPB	S	b	1, 1	171, 31	–	–
71	LPB	N	b	4, 4	342, −35	90	10
71	LPB	$N+S$	b	4, 5	346, −34	71	9
72	LPB	H	h	1, 1	183, 04	–	–
75	RB	S	b	2, 3	144, 39	27	–
75	RB	N	b	4, 5	018, −24	45	12
75	RB	$N+S$	b	6, 8	181, 32	8	20
76	NB	$S(N)$	b	5, 6	165, 28	23	14
80	NB	N	b	1, 1	333, −24	–	–
81	NB	H	h	2, 2	158, −01	8	–
82	RB	S	b	1, 1	146, 19	–	–
82	RB	$H(N)$	h	5, 5	157, −08	22	17
85	RB	S	b	9, 10	183, 37	292	3
87	NB/RB	$H(N)$	h	7, 7	149, −07	16	16
89	RB	H	h	2, 2	176, −13	–	–

Formation designations are as follows: NB = North Brook, RB = Rocky Brook, LPB = Little Pond Brook, HF = Humber Falls. The three families of magnetization are described in the text; an entry in brackets means that small magnetizations whose directions cannot be accurately determined are present; where both N and S have been determined at a single site their average irrespective of sign is given. The reference plane (b for bedding or palaeohorizontal, h for present horizontal) is given in the fourth column. The number of cores (c) and specimens (s) are indicated in the central column. D, I are the declination and inclination of the mean direction, k is Fisher's precision, and α_{95} the circle of confidence ($P = 0.05$) when s exceeds 2. The procedures for determining the magnetizations are listed in this order; site number and magnetization family, for linear segments R is the removed magnetization in the interval stated, for end-points the magnetization (or average (ave.) through a range) in the interval stated, number in brackets is the number of determinations by each method. 06*S* R450–650 °C (4), ave. 500–670 °C (4); 06*H* 650 °C (3); 09*S* R450–650 °C (2), ave. 500–670 °C (2), ave. 480–618h in acid (4); 09*N* 650 °C (1); 09*H* avg. 625–670 °C (3); 13*N* R500–650 °C (1); 20*H* ave. 450–670 °C (2); 22*S* ave. 400–670 °C (2); 22*N* ave. 400–670 °C (2); 24*S* acid 338h (2); 24*H* ave. 400–600 °C (4), R400–600 °C (5); 67*N* R18h–200h (4), R350–650 °C (15), ave. 300–600 °C (4); 70*N* ave. 300–650 °C (2); 70*H* 650 °C (3); 71*S* R300–500 °C (1); 71*N* ave. 500–650 °C (4); 72*H* ave. 550–650 °C (1); 75*S* R300–500 °C (3); 75*N* ave. 550–650 °C (5); 76*S* R300–600 °C (3), ave. 550–650 (2), acid 618h (1); 80*N* 350h (1); 81*H* R400–650 °C (2); 83*S* R70–350*h* (1); 83*H* R300–600 °C (3), ave. 500–650 °C (2); 85*S* ave. 500–650 °C; 87*H* ave. 500–650 °C (6), 350h (1); 89*H* ave. 550–680 °C (2).

magnetization removed along the long second segment has the direction 182, −19, more steeply inclined than the "apparent" end-point. Such southerly-directed magnetizations with shallow usually negative inclinations are designated *H*. In this instance, the high temperature third magnetization, although present, cannot be cleanly isolated by thermal demagnetization. However, it is

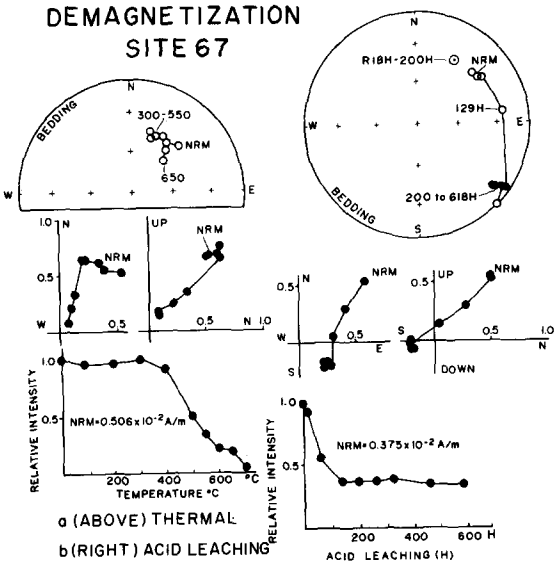


Fig. 3. Dominant N magnetization coexisting with subordinate S magnetization at collecting site 67. (a) Direction and intensity changes during thermal demagnetization together with orthogonal plots (normalized so that NRM = 1) all with respect to palaeohorizontal. The intensity at 700 °C is recorded but the direction is not, being scattered. (b) Same for acid leaching. Bedding dip is 95° (overturned) at 114° E. Solid (open) symbols on the stereograms denote downward (upward) inclination. The open symbol labelled R is the direction of the magnetization removed during leaching between 18 and 200 hours.

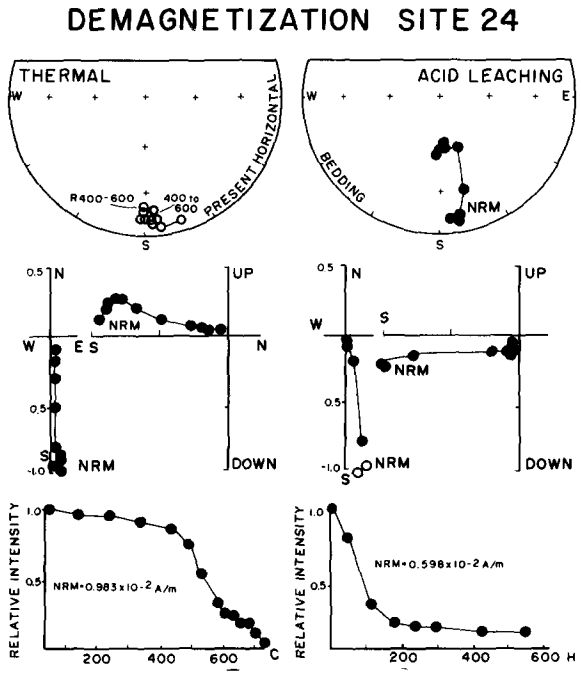


Fig. 4. Dominant H magnetization coexisting with subordinate S magnetization at site 24. Direction and intensity changes and orthogonal plots obtained by thermal demagnetization on the left, by acid leaching on the right. Bedding dip is 30° at 315°. The direction of magnetization removed during heating between 400 and 600 °C is indicated (R).

revealed by acid leaching (Fig. 4, right-hand side). Leaching up to 300 hours produces a rapid decay in intensity as the red-brown staining is dissolved and the specimen turns grey. There are corresponding linear segments on the orthogonal plots which do not march to the origin, and the magnetization being removed has the direction 172, −10, similar to that removed along the second thermal decay segment. The initial flat southerly direction rotates downward and assumes an intermediate inclination (175, +55 relative to bedding), and there is a slow steady decline in intensity with small random changes in direction, hallmarks of a true end-point. Such southerly-directed magnetizations with intermediate inclinations are designated S. S is essentially antiparallel to N. Returning to Fig. 3 we see that the indeterminate southerly-directed magnetization could be an S or an H magnetization or a mixture of both.

4. Analysis

All specimens have been demagnetized in this manner. Six gave no good linear segments because the stability spectra of the magnetizations extensively overlap and the magnetizations could not be separated. Magnetizations were satisfactorily isolated from 117 specimens (110 cores from 18 collecting sites).

The three families of directions, N, S and H, are plotted in Fig. 5 and summarized by sites in Table 2. The three families occur sporadically across the collection, areally and stratigraphically. This heterogeneity has two sources. Firstly, the magnetizations were probably acquired over a long interval of time and were preserved in an irregular fashion. For example an N magnetization may occur only in a single specimen from a collecting site whereas the remaining specimens may contain

TABLE 3

Summary of results

	Magnetization	Reference plane	Weight	n	D_m°, I_m°	k	α_{95}°	Lat.° N, long.° E ($dm°$, $dp°$)
1.	N	b	specimens	37	010.3, −37.2	25	4.8	19.6, 112.3 (5.7, 3.3)
2.	N	b	cores	31	009.3, −37.4	24	5.4	19.6, 113.3 (6.4, 3.7)
3.	N	b	sites	8	008.5, −32.8	15	15.0	22.7, 113.7 (17.0, 9.6)
4.	N	h	sites	8	007.6, −22.0	6	24.8	–
5.	S	b	specimens	41	173.9, 35.7	23	4.7	21.0, 128.7 (5.5, 3.2)
6.	S	b	cores	33	173.6, 35.5	25	5.1	21.1, 129.0 (5.9, 3.4)
7.	S	b	sites	9	171.4, 35.6	22	11.3	20.9, 131.2 (13.1, 7.6)
8.	S	h	sites	9	169.2, 19.6	18	12.6	–
9.	$N+S$	b	specimens	78	002.1, −36.7	21	3.6	20.5, 120.4 (4.2, 2.4)
10.	$N+S$	b	cores	64	001.1, −36.6	22	3.9	20.6, 121.4 (4.6, 2.7)
11.	$N+S$	b	sites	13	357.4, −35.0	19	9.7	21.6, 125.1 (11.2, 6.4)
12.	$N+S$	b	sites	8	*000.7, − 35.2*	*40*	*8.9*	*21.5, 121.8 (10.3, 6.0)*
13.	$N+S$	h	sites	8	004.8, −14.8	8	20.5	– –
14.	H	h	specimens	37	165.8, −12.4	14	6.4	45.6, 142.9 (6.5, 3.3)
15.	H	h	cores	33	164.8, −11.7	14	6.8	45.0, 144.1 (6.9, 3.5)
16.	H	h	sites	10	167.9, −14.0	18	11.7	46.9, 140.3 (12.0, 6.1)
17.	H	b	sites	10	171.0, −18.4	10	17.1	–
18.	H	h	sites	7	*167.8, − 13.3*	*37*	*10.1*	*46.5, 140.2 (10.3, 5.3)*

Headings are as for Table 2 except that n is the number of specimens, cores or sites as specified and paleopoles are given on the right. In entries 12 and 18 the means by sites for which there are more than 2 specimens are given. These are set in italics because we think they best summarize the data.

either S or H, or both. Secondly, the sampling is non-uniform for the reasons already given.

We have combined the observations in three ways, by sites, by specimens, and by cores (Table 3). Mean values are not significantly affected by weighting and the problem is to estimate fairly the errors. Our main preoccupation has been to ensure that accuracy is not overestimated so that the effects observed are real and not a consequence of the procedure adopted. It could be argued that specimens or cores, since they probably acquired their magnetizations at irregular intervals over long spans of time, provide independent determinations of the palaeofield and should therefore be given unit weight. But there could be redundancy, particularly where 5 or more magnetizations from the same family (N, S or H) have been observed from a single collecting site. A safer procedure is to give sites unit weight. Its disadvantage is that magnetizations at some sites are represented by only one determination, at others by 10 or more, so that site-means are themselves determined with variable accuracy. The compromise we suggest is to retain, for final analysis, only those sites with determinations from 3 or more specimens (Table 3). We argue that the average of three determinations from one family generally provides a reasonable estimate of that family, whereas fewer would not do so.

The N and S magnetizations are essentially antiparallel (Table 3) and probably represent reversals of the field during a time when the axis was constant. We have therefore calculated average directions for the N and S magnetizations irrespective of sign. H magnetizations are at a large angle to the N-S axis (Table 3) and have been averaged separately. Unit weight is given to those sites for which three or more determinations are present in entries 12 and 18 of Table 3, and we consider these to be a reasonable summary of our data. In these entries, errors almost certainly will not be underestimated, in fact they are likely to be overestimated. The site-means used are plotted in Fig. 5d and e.

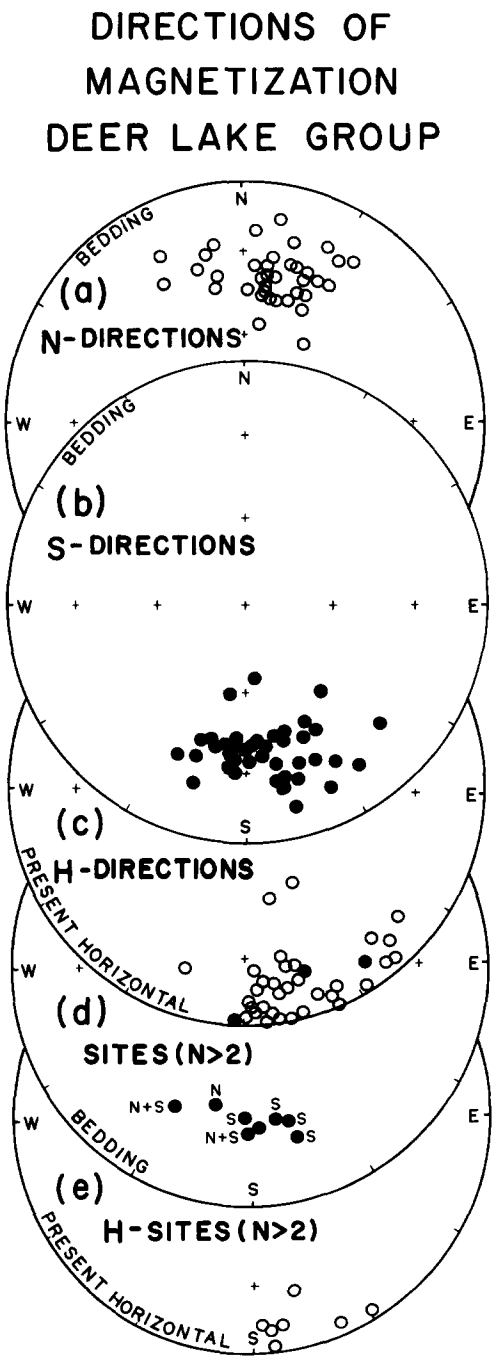


Fig. 5. (a) (b) and (c) Directions of all magnetizations observed: (d) and (e) mean directions of N, S, and H magnetizations for sites represented by 3 or more determinations.

5. Discussion of results

There is no apparent difference between directions observed in the four formations. Of the 18 sites studied, 8 have N component magnetizations, 9 have S, and 10 have H. Reversals (N and S) are present in both the North Brook and Rocky Brook Formations. N (normal polarity) predominates in the Little Pond Brook Formation. Only S (reversed polarity) has been observed in the Humber Falls. The presence of normal magnetizations (N) occurring with roughly the same frequency as reversed magnetizations (S) indicates that the N and S magnetizations predate the Kiaman—that is, they are likely to be Early Carboniferous in age and therefore roughly contemporaneous with the deposition of the Deer Lake Group. The Kiaman reversed superchron extends from the Early Westphalian to the end of the Permian [13]. This conclusion is supported by the fact that the N and S directions separately are grouped better after than before correction for tilt, especially for the N directions where the dip contrasts are greater (Table 3). Ignoring those site-results based on fewer than 3 observations, the site-means are much better grouped after ($k = 40$) than before ($k = 8$) tilt-correction. The $N + S$ magnetization predates folding. The mean N and S directions are antiparallel within errors. It is reasonable to assume therefore that N and S were acquired more or less contemporaneously, probably during diagenesis during which the palaeofield reversed at least once. The site-mean directions of the S and N magnetizations have noticeably greater spread in declination than inclination (Fig. 5d). This could have been caused by small imperfections in tilt corrections [14], or by small tectonic rotations about vertical axes among sites.

The H magnetization is always reversed in direction. The grouping is somewhat better before than after correction for tilt, but the dip contrast at H sites is too small for an adequate test. As we show below, the H magnetization is very probably an early Kiaman (latest Carboniferous or early Permian) overprint, which post-dates the folding and which was acquired some 50 Ma after N and S. Textural studies [15] of the iron minerals have shown that Deer Lake rocks commonly contain

secondary hematite, probably related to groundwater circulation. The required oxidizing conditions could have persisted to great depths as the water-table fell during the widespread desert conditions at the centre of Pangaea where Newfoundland was situated at the end of the Palaeozoic [16]. The alternative, that it was acquired during uplift and cooling following deep burial, is less plausible, because the rocks have been buried only up to present sediment thicknesses and heating even to 150 °C seems very unlikely (R. Hyde, personal communication).

6. Comparison with other data

Comparisons with other results from North America are made by transforming the mean directions first into palaeopoles and then into their corresponding palaeolatitudes and palaeosouth (because all are reversed) directions. The *H* magnetizations are compared with the early Kiaman field observed for cratonic North America in Fig. 6. The palaeolatitude for *H* (7 ± 5 °N) is not different from that expected and palaeosouth is also in good accord with the Kiaman grid.

Comparison with results from middle and Early Carboniferous (Tournaisian to Namurian) of the Canadian Appalachians are shown in Fig. 7. The pre-folding magnetization in the Deer Lake Group is statistically indistinguishable from the pre-folding magnetization observed by Kent [17] in the Tournaisian [18] Terenceville Formation of the Avalon Platform of eastern Newfoundland. The palaeolatitude of the Terenceville has been calculated from Table 1 of Kent [17] eliminating site F, which we suspect has been affected by Kiaman overprinting. The result is as follows: 8 sites, 180°, 31°, $k = 45$, $\alpha_{95} = 8$°, palaeopole 25 °N, 125 °E, $dm = 9$°, $dp = 5$°, palaeolatitude 17 ± 5°. In Kent [17] a mean inclination of 28° include site F and a palaeolatitude of 13° is given. The latter is incorrect, it should have been 15°. We prefer our analysis because it eliminates all suspicion of Kiaman overprinting, but whichever result is used (ours 17 ± 5° or the corrected result from Kent [17] 15 ± 6°) the eastern (Avalon) and western Newfoundland platforms (Fig. 2, inset) had statistically identical palaeolatitudes in the Early

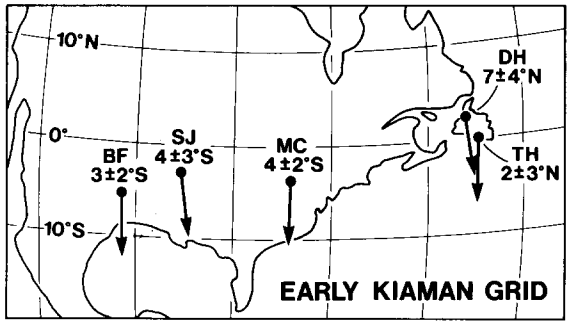


Fig. 6. Palaeolatitudes obtained from Early Carboniferous rocks compared with early Kiaman palaeolatitudes. The Early Kiaman grid is calculated for a palaeopole 43 °N, 126 °E which is the mean for the interval 265–295 Ma [32]; *BF* = Barnett Formation (Mississippian) [25]; *SJ* = St. Joe limestone (Tournaisian) [26]; *MC* = Mauch Chunk Formation (late Mississippian) [27]. Palaeolatitudes from Kiaman overprints from Newfoundland Early Carboniferous rocks are also shown: *DH* = Deer Lake Group *H* magnetizations; *TH* = Terenceville Formation overprint [17].

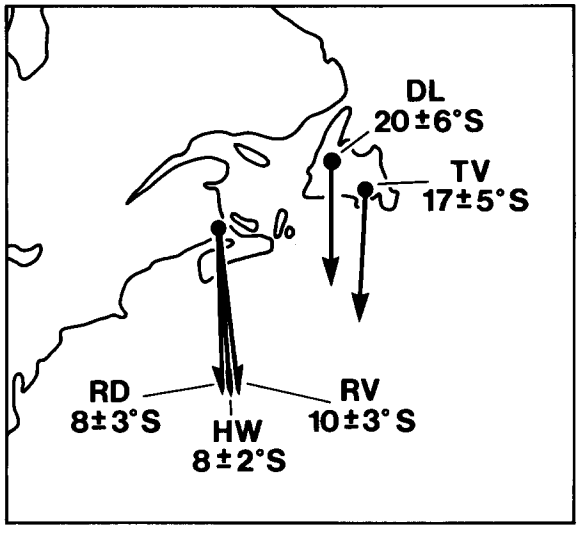


Fig. 7. Palaeolatitude of middle and Early Carboniferous rocks of Appalachians compared. *DL* = Deer Lake Group (Viséan to possibly early Namurian) *N-S* magnetization (this paper); *TV* = Terenceville Formation (Tournaisian) prefolding magnetization [17]; *HW* = Hopewell Group (Namurian) [20,21]; *RV* = Riversdale Group later magnetization (Namurian/early Westphalian) [22]; *RD* = Riversdale later magnetization (Namurian/early Westphalian) [22].

Carboniferous. Hence it is unlikely that they have moved by large distances relative to one another since then. These two results are also in agreement with the palaeolatitude of 22°S reported in abstract by Opdyke [19] from the Westphalian Barachois Group of southwestern Newfoundland. Admittedly there is an age difference between the Viséan Deer Lake Group and the Tournaisian Terenceville Formation but the good agreement, whilst it allows some relative movement, precludes very large displacement.

Palaeolatitudes from the New Brunswick–Nova Scotia Carboniferous sedimentary basin range between about 7 and 10°S, significantly lower than those from Carboniferous rocks of Newfoundland (Fig. 7). The oldest beds studied from the mainland are Namurian (Hopewell Group [20,21]) and Namurian to earliest Westphalian (Riversdale Group, [22]) in age, and are younger than the Newfoundland Carboniferous rocks that have been studied. The difference could be reasonably explained by a systematic time variation of palaeolatitude, by a northerly motion of and North America as a whole during the middle Carboniferous [23].

Difficulties are immediately encountered when comparing results for the Early Carboniferous from the Appalachians with those for cratonic North America because the latter are few (3 all told) and of variable reliability [23]. Furthermore, as we have just seen, palaeolatitudes probably changed during the Carboniferous so that comparisons between the craton and the orogenic margin should be made between results from roughly contemporaneous rocks. The data to make such comparisons properly are not available.

A further important problem is the possibility that cratonic rocks have been subject to widespread Kiaman overprinting [23,24]. This is a very real possibility because the magnetizations of the Mississippian Barnett Formation [25], the Tournaisian St. Joe Limestone [26] and the Viséan Mauch Chunk Formation [27], when translated into palaeolatitudes and palaeosouth directions, are statistically indistinguishable from those obtained from latest Carboniferous and Early Permian (Kiaman) rocks of cratonic North America [23], and from the overprint *H* magnetizations of the Deer Lake Group and the Terenceville Formation (Fig. 6). In none of the determinations from Early Carboniferous cratonic rocks have Kiaman overprints been rigorously isolated, yet they are ubiquitous in the Appalachians and elsewhere in the world [23,24]. Moreover, normal magnetizations, characteristic of middle and Early Carboniferous rocks elsewhere, are very rare. Indeed they comprise only a few per cent of the samples studied from rocks of this age from cratonic North America. Yet in the Deer Lake Group, normal and reversed magnetizations occur with equal frequency, as they do in mid-Carboniferous rocks of the Maritime Provinces of Canada [23] and elsewhere in the world [13] in rocks of this age. There is no good reason why the Early Carboniferous cratonic rocks of the U.S.A. should be exceptional with regard to the occurrence of reversals. Finally we note that the data from the Mauch Chunk Formation are better grouped after than before correction for bedding tilt [27]. This, however, does not necessarily imply the absence of early Kiaman overprinting because the Alleghanian folding occurred in the latest Carboniferous or earliest Permian and hence could post-date the acquisition of an early Kiaman overprint. A plausible mechanism for this has recently been documented by Hodych et al. [28]. They showed that Silurian folded beds in the southern Appalachians acquired their very stable magnetization by the transformation of goethite to hematite at about 80°C during burial in the late Carboniferous just prior to Alleghanian folding but long after deposition. A similar process could be responsible for the southerly-directed magnetization most commonly observed in the Mauch Chunk Formation.

Our new data strengthen these arguments. The Deer Lake Group rests unconformably on and contains clasts of Grenvillian basement, and hence is undeniably part of cratonic North America. Frequent normal polarities occur within it. The pre-folding *N-S* magnetization and the Kiaman overprint have been separated. All the criteria for the acceptance of the *N-S* magnetization as a valid determination of the Early Carboniferous palaeofield have been satisfied, which is not so for other studies of Early Carboniferous cratonic rocks. Hence we suggest that the Deer Lake Group *N-S*

magnetization is the only currently valid determination of the Early Carboniferous palaeofield for cratonic North America. This would indicate that the southern half of North America was situated well within the southern hemisphere, 20–30°S, and this is consistent with the palaeoclimatic evidence [29]. Other determinations from Early Carboniferous rocks of North America are suspect because the Kiaman overprint has not been clearly identified and separated by laboratory analysis, because normal polarity magnetizations are untypically rare, and because the directions are statistically identical to the much younger Kiaman palaeofield (Fig. 6). Finally we note that normal polarities with inclinations of −39 and −42° corresponding to palaeolatitudes of 22 and 24°S respectively have been observed from two specimens of the Mauch Chunk Formation [30]. These specimens possess univectorial magnetization, being unchanged by heating up to 650°C. The basis is insufficient for firm conclusions, nevertheless these two indisputably single-component magnetizations from the Mauch Chunk do yield a southerly palaeolatitude consistent with that which we have observed in the essentially contemporaneous Deer Lake Group.

7. Tectonic significance

The *N-S* magnetization of the Deer Lake Group of western Newfoundland has much higher inclinations and corresponding palaeolatitudes than those obtained from Early Carboniferous rocks of cratonic North America. The Deer Lake observations however agree well with Early Carboniferous results from eastern Newfoundland, and differences with results from somewhat younger Carboniferous rocks the Maritime Provinces are no more than can reasonably be expected from the age spread of the units studied. It seems to us therefore that the suggestion [6] that "Acadia" was displaced northwards by about 1500 km relative to cratonic North America in the mid-Carboniferous is no longer supported by the palaeomagnetic data. Widespread Kiaman overprinting of the cratonic record seems more likely to have been responsible for the apparent discrepancy between the craton and "Acadia". This analysis does not discount the possibility of movement of "Acadia" relative to cratonic North America in the earliest Carboniferous latest Devonian or earlier times. These questions will be addressed in a second paper [31]. The analysis does, however, tell us that it is very unlikely that such motion occurred during the Viséan or later; it tells us that there is no currently available palaeomagnetic evidence to support the idea that "Acadia" was situated 1500 km to the south in the Early Carboniferous and that it has moved north to its present position later in the Carboniferous. Of course smaller displacements of several hundred kilometres would not be detected by the palaeomagnetic method.

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